Liquidity Monitoring and Measurement

Stress Testing

Liquidity stress testing is performed for each of Citi's major entities, operating subsidiaries and/or countries. Stress testing and scenario analyses are intended to quantify the potential impact of a liquidity event on the balance sheet and liquidity position, and to identify viable funding alternatives that can be utilized. These scenarios include assumptions about significant changes in key funding sources, market triggers (such as credit ratings), potential uses of funding and political and economic conditions in certain countries. These conditions include expected and stressed market conditions as well as Company-specific events.

Liquidity stress tests are conducted to ascertain potential mismatches between liquidity sources and uses over a variety of time horizons (overnight, one week, two weeks, one month, three months, one year) and over a variety of stressed conditions. Liquidity limits are set accordingly. To monitor the liquidity of an entity, these stress tests and potential mismatches are calculated with varying frequencies, with several tests performed daily.

Given the range of potential stresses, Citi maintains a series of contingency funding plans on a consolidated basis and for individual entities. These plans specify a wide range of readily available actions for a variety of adverse market conditions or idiosyncratic stresses.

Short-Term Liquidity Measurement: Liquidity Coverage Ratio (LCR)

In addition to internal measures that Citi has developed for a 30-day stress scenario, Citi also monitors its liquidity by reference to the LCR, as calculated pursuant to the U.S. LCR rules.

Generally, the LCR is designed to ensure that banks maintain an adequate level of HQLA to meet liquidity needs under an acute 30-day stress scenario. The LCR is calculated by dividing HQLA by estimated net outflows over a stressed 30-day period, with the net outflows determined by applying prescribed outflow factors to various categories of liabilities, such as deposits, unsecured and secured wholesale borrowings, unused lending commitments and derivatives-related exposures, partially offset by inflows from assets maturing within 30 days. Banks are required to calculate an add-on to address potential maturity mismatches between contractual cash outflows and inflows within the 30-day period in determining the total amount of net outflows. The minimum LCR requirement is 90% effective January 2016, increasing to 100% in January 2017.

The table below sets forth the components of Citi's LCR calculation and HQLA in excess of net outflows as of the periods indicated:

In billions of dollars	**Dec. 31, 2015**	Sept. 30, 2015	Dec. 31, 2014
HQLA	**$ 378.5**	$ 398.9	$ 412.6
Net outflows	**336.5**	355.6	368.6
LCR	**112%**	112%	112%
HQLA in excess of net outflows	**$ 42.0**	$ 43.3	$ 44.0

As set forth in the table above, Citi's LCR was unchanged both year-over-year and quarter-over-quarter, as the reduction in Citi's HQLA was offset by a reduction in net outflows, reflecting reductions in Citi's long-term debt and short-term borrowings.

Long-Term Liquidity Measurement: Net Stable Funding Ratio (NSFR)

For 12-month liquidity stress periods, Citi uses several measures, including its internal long-term liquidity measure, based on a 12-month scenario assuming deterioration due to a combination of idiosyncratic and market stresses of moderate to high severity. It is broadly defined as the ratio of unencumbered liquidity resources to net stressed cumulative outflows over a 12-month period.

In addition, in October 2014, the Basel Committee on Banking Supervision (Basel Committee) issued final standards for the implementation of the Basel III NSFR, with full compliance required by January 1, 2018. Similar to Citi's internal long-term liquidity measure, the NSFR is intended to measure the stability of a banking organization's funding over a one-year time horizon. Pursuant to the Basel Committee's final standards, the NSFR is calculated by dividing the level of a bank's available stable funding by its required stable funding. The ratio is required to be greater than 100%. Under the Basel Committee standards, available stable funding primarily includes portions of equity, deposits and long-term debt, while required stable funding primarily includes the portion of long-term assets which are deemed illiquid. The U.S. banking agencies have not yet proposed the U.S. version of the NSFR, although a proposal is expected during 2016.